

SEREN

Every child is a star. Every adult deserves a second chance.

MASTER RESEARCH & TECHNICAL BLUEPRINT — LAUNCH-READY EDITION

SEREN: AI-Powered Universal Screening & Intervention Platform for 80+ Neurodevelopmental, Communication, Emotional, and Confidence-Based Conditions Across the Full Human Lifespan — Detection Technology, Signal Architecture, Clinical Framework, and Product & Deployment Blueprint

80+	400+	12	95-99%	5-60+
Conditions Covered	Detection Signal Points	Detection Modalities	Target Accuracy	Ages Served

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Abstract

This paper presents the complete scientific, technological, and commercial foundation for SEREN — an AI-powered, offline-first, universal screening and intervention platform for 80+ neurodevelopmental, communication, emotional, sensory, and confidence-based conditions across the full human lifespan, from children aged 5 through adults aged 60 and above. This edition expands the original framework to include: (1) the full epidemiological burden across all 80+ conditions in India and globally; (2) the neuroscience, behavioural manifestations, and clinical profiles for each condition, organised by life stage — child, teen, and adult; (3) a comprehensive mobile-first detection technology stack combining computer vision, handwriting analysis, speech and audio machine learning, natural language processing, passive sensor fusion, and gamified hidden assessment, designed to achieve 95-99% screening accuracy without any specialist hardware; (4) a condition-by-condition signal catalogue of 400+ individual detection signals; (5) an evidence-based, age-adapted treatment and intervention architecture; and (6) a complete product, engineering, and go-to-market blueprint describing exactly how SEREN is built, validated, and deployed as a launch-ready product.

SEREN is designed around three structural innovations. First, it is architected to screen simultaneously across learning disorders, speech and language conditions, attention and processing profiles, anxiety and emotional conditions, sensory and motor patterns, silent neurodivergent profiles, and adult insecurity and confidence deficits within a single adaptive session, rather than requiring separate tools for each domain. Second, it targets near-clinical accuracy using only a standard mobile device — no specialist hardware, no clinical setting, and no trained administrator required. Third, it is built fully offline-first, so it can be deployed in low-connectivity and resource-constrained environments across India and other emerging markets.

The underlying need is substantial. Screening research places learning-disability prevalence in the Indian school-age population in the range of 10-12%, translating to tens of millions of children carrying at least one undetected neurodevelopmental condition at any given time. These conditions do not resolve at graduation — they persist into adulthood as unspoken barriers to career progression, communication confidence, and mental wellbeing. SEREN is designed to interrupt that trajectory at the earliest possible point, and to give adults who were never screened as children a pathway to understanding and addressing conditions that have quietly shaped their lives.

This edition further aligns SEREN's architecture with India's National Education Policy (NEP) 2020, which mandates universal Foundational Literacy and Numeracy (FLN) by the end of Grade 3 through the NIPUN Bharat mission, and which explicitly calls for early identification and support of children with learning difficulties within regular classrooms. SEREN is positioned as a direct, technology-enabled instrument for operationalising this policy mandate at scale.

Keywords

Dyslexia, Dyscalculia, ADHD, Dysgraphia, Selective Mutism, Social Anxiety, Stuttering, Word-Finding Difficulty, Processing Speed Disorder, Sensory Processing Disorder, Body Image Insecurity, Imposter Syndrome, Rejection Sensitivity, AI Screening, Eye-Tracking, Handwriting CNN, Speech ML, Webcam

rPPG, Passive Sensor Fusion, Gamified Intervention, NEP 2020, Foundational Literacy and Numeracy, India, SEREN, Neurodiversity, Adult Intervention, EdTech, HealthTech.

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SECTION 1 — The Landscape: Epidemiology of 80+ Undetected Conditions

1.1 India: Scale of the Undetected Child and Adult

India's school-age population stands at approximately 250 million children between ages 5 and 14 — the largest single cohort of school-age children in the world. A systematic review published in the Indian Journal of Psychiatric Nursing (2022) found a pooled Learning Disability prevalence of 10.70% (95% CI: 7.10%-14.3%) across Indian studies. A South Indian study (Archives of Mental Health, 2023) found dyslexia at 11.2%, dysgraphia at 12.5%, and dyscalculia at 10.5%. Extrapolated conservatively across the national school-age population, this places the number of Indian children carrying at least one undetected neurodevelopmental condition at well over three crore. Beyond childhood, India has an estimated tens of millions of adults living with undiagnosed conditions that originated in childhood — professionals who still stutter under pressure, employees who quietly avoid writing tasks, and adults who read slowly in private while appearing entirely capable in every other respect.

SEREN's founding principle is that the average Indian child with dyslexia is not identified until several years after entering school — by which point years of academic mislabeling, reduced confidence, and mismatched instruction have already shaped their trajectory. This is not primarily a gap in awareness; it is a gap in scalable, affordable screening infrastructure. SEREN is built to be that infrastructure — for children and for the adults they grow into.

1.2 Global Prevalence — Representative Conditions Across All 80+

The table below summarises indicative global prevalence for a representative cross-section of the 80+ conditions SEREN screens for, spanning learning, attention, speech, anxiety, emotional, sensory, silent/neurodivergent, and insecurity-based categories. Full per-condition detail, including India-specific estimates and adult persistence patterns, is provided in Section 2.

Condition	Category	Global Prevalence
Dyslexia	Learning Disorders	5-20% globally
Dyscalculia	Learning Disorders	5-7% globally
Dysgraphia	Learning Disorders	5-20% globally
ADHD (all presentations)	Attention & Processing	7.2% globally
Auditory Processing Disorder	Attention & Processing	2-5% globally
Stuttering	Speech & Language	1% globally
Word-Finding Difficulty	Speech & Language	5-7% globally
Generalised Anxiety Disorder	Anxiety Disorders	6.5% children; 3.7% adults
Social Anxiety Disorder	Anxiety Disorders	7-13% lifetime
Selective Mutism	Anxiety Disorders	0.47-0.76% globally
Exam / Test Anxiety	Anxiety Disorders	10-40% of students
Childhood Depression (masked)	Emotional & Behavioural	2-3% of children

Condition	Category	Global Prevalence
Emotional Dysregulation	Emotional & Behavioural	~8% globally
Masking and Camouflaging	Silent Profiles	~5% globally
High-Functioning Autism (masked)	Silent Profiles	1-2% globally
Sensory Processing Disorder	Sensory & Motor	5-16% globally
Body Image Insecurity	Insecurities	~20% of teens
Imposter Syndrome	Insecurities	~70% of professionals
Rejection Sensitivity (RSD)	Insecurities	~30% in ADHD; ~15% general
Workplace / Career Insecurity	Insecurities	~35% of workforce

A complete condition-by-condition prevalence table, cross-referenced against India-specific estimates and adult persistence rates, appears in Section 2 alongside the full clinical profile for each condition.

1.3 The Adult Burden: Conditions That Follow You Forever

Dyslexia, ADHD, anxiety, stuttering, and the majority of conditions in this catalogue do not disappear at age 18. They are lifelong neurological and psychological profiles that continue to shape professional and personal outcomes long after formal schooling ends. Research on adults with unidentified ADHD documents measurably lower average annual income relative to peers (Fayyad et al., 2007), alongside disproportionate rates of university dropout, career plateauing, relationship instability, and elevated use of substances as an unrecognised coping mechanism. A 2024 study published via PMC demonstrated that a structured web-based linguistic intervention delivered over 24 sessions of 15-20 minutes each produced measurable literacy improvements in adults aged 16-30 with dyslexia — confirming that neuroplasticity supporting meaningful intervention persists well into adulthood. SEREN's adult module is directly modelled on this evidence base.

A capable, intelligent adult who has never had a name for the difficulty they have carried since childhood is not a rare case — it is a structural gap repeated across millions of working adults. SEREN's screening and intervention architecture is deliberately designed to reach this population as directly as it reaches children in classrooms.

1.4 Systemic Gap Analysis and the NEP 2020 Alignment Opportunity

India's National Education Policy (NEP) 2020 designates Foundational Literacy and Numeracy (FLN) as the highest immediate priority of the Indian school system, stating explicitly that the rest of the policy “will become relevant to our students only if this most basic learning requirement is first achieved.” The policy established the National Mission on Foundational Literacy and Numeracy, subsequently operationalised as the NIPUN Bharat mission, targeting universal FLN by the end of Grade 3. NEP 2020 further commits to equitable and inclusive education, with explicit provisions for identifying and supporting children with learning difficulties within mainstream classrooms rather than routing them into separate systems.

This creates a direct and current policy opening. NIPUN Bharat requires states to identify foundational learning gaps early and track them systematically, yet no nationally scaled, zero-cost, offline-capable screening instrument currently exists to do this at the level of individual learning disorders and

processing differences. SEREN is positioned to function as exactly this instrument: a Grade 1-3 deployable, teacher-facing, regional-language screening layer that gives schools and state FLN missions the individual-level diagnostic granularity that population-level literacy surveys cannot provide. This is a structural advantage for SEREN as a startup — its product roadmap is not competing against policy, it is built to fulfil an explicit unmet requirement of a live national mission.

Gap	Current Reality	SEREN's Response
Access	Specialist child-development assessment is scarce and costly outside major metro centres, with long waitlists at public referral institutions.	Free, on-device, school- or self-deployed screening with zero travel requirement.
Awareness	Minimal structured training on specific learning disorders in teacher education programmes; limited regional-language vocabulary for common conditions.	Teacher- and parent-facing reports generated in Hindi, Marathi, and other Indian languages using plain, non-clinical language.
Infrastructure	No nationally validated, zero-cost, offline-capable, multi-language screening tool exists at scale; most global tools are English-only, internet-dependent, and priced per-student.	Offline-first architecture; multi-language support; near-zero marginal cost per student; runs on entry-level Android devices.
Comorbidity	A meaningful share of children with one learning difficulty also carry a co-occurring emotional condition; single-condition tools miss the combined picture.	80+ conditions screened within a single adaptive session; comorbidity patterns detected automatically via multimodal fusion.
Adult neglect	No structured system exists for adults to discover childhood-onset conditions that were never identified; many adults have no name for a lifelong struggle.	Dedicated adult screening and intervention module with short daily sessions designed for working adults.
Policy execution gap	NEP 2020 and NIPUN Bharat mandate universal FLN identification, but implementation currently relies mostly on population-level literacy surveys rather than individual diagnostic screening.	SEREN provides individual-level, condition-specific screening data that state FLN missions can use directly for classroom-level intervention planning.

SECTION 2 — The Conditions: Complete Neuroscience, Profiles & Signal Catalogue

This section organises all 80+ conditions SEREN screens for into four life-stage groupings — Child (ages 5-12), Teen (ages 13-19), Adult (ages 18-60+), and All-Ages / cross-cutting conditions — reflecting how each condition most typically presents and is first detected. Flagship conditions within each group are presented with a full clinical profile: neurological basis, behavioural manifestations, and the specific SEREN detection signals used to flag them. The remaining conditions in each group are presented in summary table form, with full signal-level detail available in the SEREN technical appendix maintained alongside this paper.

2.1 Child Conditions (Ages 5-12) — Learning, Attention, Speech & Sensory

Childhood is the highest-leverage window for detection: conditions identified and supported early carry substantially better lifetime trajectories than those first recognised in adolescence or adulthood. This group covers the classic learning disorders, attention and processing differences, developmental speech and language conditions, and sensory-motor patterns that most commonly present during primary school years.

Dyslexia

Category: Learning Disorders | **Global Prevalence:** 5-20% global | **SEREN Target Accuracy:** 95-99%

Neurological / Psychological Basis: Phonological processing deficit with reduced activation in the left-hemisphere reading network (temporoparietal junction, visual word form area, Broca's area); heritability estimated at 50-60%; associated genes include DYX1C1, DCDC2, and KIAA0319.

Manifestations:

- Reading significantly below grade level despite adequate instruction and typical general ability
- Letter reversals (b/d, p/q confusion) persisting beyond the age this is developmentally expected
- Slow, laborious oral reading with frequent omissions and substitutions
- Difficulty with phoneme blending, rhyme, and syllable segmentation
- Inconsistent spelling that violates common phonetic rules
- Reading fatigue and avoidance of reading tasks
- Strong listening comprehension relative to markedly weaker reading comprehension
- In adults: slow professional reading, spelling errors in written communication, avoidance of writing-heavy tasks

SEREN Detection Signals:

- Webcam eye-tracking: fixation duration, regression frequency, saccade amplitude
- Rapid Automatized Naming (RAN) latency across letters, digits, objects, and colours
- Phoneme segmentation accuracy and response time

- Handwriting CNN analysis: letter-reversal rate and stroke-direction pattern
- Oral reading fluency measured in words-per-minute via on-device speech recognition
- Sight-word recognition latency on timed flashcard tasks
- Rhyme-detection accuracy on an audio discrimination task

Scientific Basis: *International Dyslexia Association (2023); Snowling & Hulme (2011); Vellutino et al. (2004); Zaibi & Bezine (2024).*

Dyscalculia

Category: Learning Disorders | **Global Prevalence:** 5-7% global | **SEREN Target Accuracy:** 88-93%

Neurological / Psychological Basis: Impaired core “number sense” linked to atypical intraparietal sulcus activation and a right-hemisphere parietal network deficit; distinct from maths anxiety, though the two co-occur in a substantial minority of cases.

Manifestations:

- Cannot automatically recall basic arithmetic facts despite repeated practice
- Difficulty with place value and number magnitude comparison
- Finger-counting persisting well beyond the developmentally expected age
- Confusion with mathematical symbols and operations
- Poor comprehension of time, money, and measurement concepts
- Elevated maths anxiety, including physical symptoms before maths-related tasks

SEREN Detection Signals:

- Dot enumeration (subitizing) accuracy and speed for arrays of 2-8 dots
- Number-magnitude comparison latency
- Arithmetic fact-retrieval speed under timed conditions
- Number-line estimation deviation score
- Approximate Number System (ANS) task accuracy
- Place-value understanding task using concrete-representational-abstract sequencing

Scientific Basis: *Butterworth (2010); Dehaene (1997); Wilson et al. (2006); Landerl et al. (2004).*

Dysgraphia

Category: Learning Disorders | **Global Prevalence:** 5-20% global | **SEREN Target Accuracy:** 95-99%

Neurological / Psychological Basis: Three recognised subtypes — phonological (grapheme-phoneme mapping), surface (irregular spelling patterns), and motor (overlapping with coordination difficulty) — with neural correlates overlapping both the dyslexia network and fine-motor coordination pathways.

Manifestations:

- Illegible handwriting inconsistent with effort and general ability

- Cramped pencil grip and visible fatigue during writing tasks
- Inconsistent letter sizing, spacing, and baseline alignment
- Difficulty organising ideas in writing despite clear verbal expression
- Writing speed well below grade-level expectations
- Avoidance of writing tasks and frequently incomplete written work

SEREN Detection Signals:

- Touchscreen letter-formation CNN measuring stroke velocity, acceleration, and jerk coefficient
- Stylus pressure proxy from capacitive touch-area data
- Letter-height variability and inter-letter spacing consistency analysis
- Writing speed in characters per minute against age-normed benchmarks
- Baseline-drift detection across a multi-sentence copying task
- Direct stylus pressure data where supported hardware is available

Scientific Basis: *Berninger et al. (2008); Zaibi & Bezine (2024); Rangasrinivasan et al. (2025).*

ADHD - Inattentive Type

Category: Attention & Processing | **Global Prevalence:** 7.2% global | **SEREN Target Accuracy:** 90-95%

Neurological / Psychological Basis: Associated with reduced prefrontal cortex, basal ganglia, and cerebellar volume, and dopaminergic/noradrenergic dysregulation; heritability estimated at 75-80%. Most frequently under-identified in girls and in children with high general ability.

Manifestations:

- Difficulty sustaining attention; frequently drifts off mid-task
- Fails to follow through on multi-step instructions; work left incomplete
- Loses materials and appears forgetful in daily routines
- Easily distracted by both external and internal stimuli
- Struggles with task organisation and realistic time estimation
- In adults: chronic disorganisation, procrastination, and missed deadlines

SEREN Detection Signals:

- Continuous Performance Test: miss rate over a sustained multi-minute attention task
- Vigilance-decrement index measuring performance drop across the session
- Eye-gaze deviation during instruction screens
- Omission-error rate on a target-detection task
- Passive signal: session interruption frequency (device backgrounded, app exited)
- Response-inconsistency index (standard deviation of reaction time across trials)

Scientific Basis: *Conners (2000); Barkley (2015); Willcutt et al. (2005).*

Stuttering (Developmental Fluency Disorder)

Category: Speech & Language | **Global Prevalence:** 1% global | **SEREN Target Accuracy:** 90-95%

Neurological / Psychological Basis: Neurological timing disruption in speech motor planning, implicating basal ganglia-thalamo-cortical circuitry; heritability approximately 70%; roughly four-fifths of childhood-onset cases resolve spontaneously, with the remainder persisting — including into professional adulthood.

Manifestations:

- Syllable or sound repetitions
- Sound prolongations
- Speech blocks, where airflow or voicing stops entirely
- Secondary behaviours such as eye blinking, head jerks, or lip tension
- Avoidance vocabulary — deliberately choosing easier words to avoid feared sounds
- Anticipatory anxiety before speaking situations
- Career and relationship limitations that can persist across a lifetime without intervention

SEREN Detection Signals:

- Syllable-repetition detection via a custom disfluency audio classifier
- Sound-prolongation duration via spectral analysis
- Block detection: intra-word silence exceeding a defined threshold
- Secondary-behaviour detection: facial landmark tracking correlated with speech timing
- Avoidance-vocabulary flagging via lexical complexity analysis relative to predicted word choice
- Fluency variance between low-pressure and simulated-audience speaking conditions
- Anticipatory-anxiety proxy: response delay before high-frequency trigger consonants

Scientific Basis: *Guitar (2014); Bloodstein & Ratner (2008); Chang et al. (2008).*

Selective Mutism (SM)

Category: Anxiety Disorders | **Global Prevalence:** 0.47-0.76% global | **SEREN Target Accuracy:** 82-88%

Neurological / Psychological Basis: Consistent failure to speak in specific social settings such as school, despite typical speech at home; anxiety-based rather than oppositional in origin.

Manifestations:

- Speaks normally at home; mute or near-mute at school
- Communicates through gestures, pointing, or writing
- Visible distress or freezing when expected to speak
- Frequently misidentified by adults as shyness, stubbornness, or an autism-spectrum trait
- Without early intervention, can develop into chronic social anxiety in adulthood

SEREN Detection Signals:

- Speech-initiation latency during prompted speaking tasks
- Whisper-versus-normal-voice detection across session contexts
- Situational speech mapping across solo, paired, and group task conditions
- Freeze-response detection: motor-activity pause when speech is expected
- Speech-latency differential between familiar and unfamiliar interaction partners

Scientific Basis: *McHolm et al. (2005); Cohan et al. (2008); Kehle et al. (1998).*

Childhood Depression (Masked)

Category: Emotional & Behavioural | **Global Prevalence:** 2-3% children | **SEREN Target Accuracy:** 80-86%

Neurological / Psychological Basis: Dysphoric mood and cognitive slowing associated with HPA-axis dysregulation and reduced prefrontal cortical volume; frequently masked in children as simple quietness rather than recognised distress.

Manifestations:

- Persistent low energy and reduced interest in previously enjoyed activities
- Negative self-talk
- Psychomotor slowing: slowed movement and speech
- Concentration and memory difficulty
- Physical complaints without an identifiable medical cause

SEREN Detection Signals:

- Energy-level drop across the session, measured as performance deterioration over time
- Sentiment analysis of self-referential language for negative content
- Anhedonia proxy: reduced engagement in reward-based game tasks
- Psychomotor slowing: elevation across all response-latency measures
- Future-negative language patterns detected via NLP

Scientific Basis: *Kovacs (1985); Weissman et al. (1999); Thapar et al. (2012).*

Additional Child-Presenting Conditions (Summary)

Condition	Category	Prevalence	Target Accuracy
Dyspraxia / Developmental Coordination Disorder (DCD)	Learning Disorders	5-6% global	85-90%
Irlen Syndrome / Visual Stress (Meares-Irlen Syndrome)	Learning Disorders	12-15% general; up to 40% dyslexia-comorbid	72-80%
Non-Verbal Learning Disability (NVLD)	Learning Disorders	~3-4% global	75-82%
ADHD - Hyperactive/Impulsive Type	Attention & Processing	7.2% global	88-93%

Condition	Category	Prevalence	Target Accuracy
ADHD - Combined Type	Attention & Processing	~50% of ADHD cases	90-95%
Processing Speed Disorder	Attention & Processing	~5% global	88-93%
Auditory Processing Disorder (APD)	Attention & Processing	2-5% global	82-88%
Working Memory Deficit (non-ADHD)	Attention & Processing	~10% of school population	78-85%
Expressive Language Disorder	Speech & Language	3-5% global	83-88%
Receptive Language Disorder	Speech & Language	2-3% global	82-87%
Pragmatic Language / Social Communication Disorder (SCD)	Speech & Language	2-4% global	78-85%
Childhood Apraxia of Speech (CAS)	Speech & Language	0.1-0.2% global	80-87%
Phonological Disorder (Speech Sound Disorder)	Speech & Language	2-25% (age-dependent)	85-90%
Separation Anxiety Disorder	Anxiety Disorders	4-5% children	78-84%
School Refusal	Anxiety Disorders	2-5% children	80-86%
Specific Phobia	Anxiety Disorders	5-10% children	76-83%
Oppositional Anxiety-Driven Behaviour	Emotional & Behavioural	~3% children	75-81%
Social Communication Disorder (isolated)	Silent Profiles & Neurodiversity	2-4% global	78-85%
Twice-Exceptional Profile (2e)	Silent Profiles & Neurodiversity	~2% global	77-83%
Pathological Demand Avoidance (PDA) Profile	Silent Profiles & Neurodiversity	~1% global	73-79%
Sensory Processing Disorder (SPD)	Sensory & Motor	5-16% global	78-84%
Sensory Defensiveness (Tactile/Auditory Hypersensitivity)	Sensory & Motor	~10-15% children	76-82%
Vestibular Processing Difficulty	Sensory & Motor	~5-7% children	74-80%
Slow Processing of Written Instructions (Reading Fluency Lag)	Learning Disorders	~8% school population	80-86%
Orthographic Processing Deficit (Sight-Word Memory)	Learning Disorders	~4-6% school population	82-88%
Illness-Related Health Anxiety in Children (Medical Trauma)	Anxiety Disorders	~3-5% chronically ill children	72-79%
Post-Disruption Learning-Loss Anxiety	Emotional & Behavioural	~20% of affected cohorts	70-76%
Proprioceptive Processing Difficulty	Sensory & Motor	~5% children	73-79%

2.2 Teen Conditions (Ages 13-19) — Identity, Anxiety & Social Profiles

Adolescence introduces a distinct layer of conditions centred on identity formation, social comparison, and performance pressure, often overlaying earlier undetected learning or attention differences. Screening in this age band must account for masking behaviours that intensify during the teenage years as social stakes rise.

Social Anxiety Disorder (SAD)

Category: Anxiety Disorders | **Global Prevalence:** 7-13% lifetime | **SEREN Target Accuracy:** 87-92%

Neurological / Psychological Basis: Fear of negative social evaluation associated with right amygdala hyperactivation; avoidance behaviour compounds over time and typically peaks in prevalence during adolescence.

Manifestations:

- Intense fear of being watched, judged, or embarrassed
- Avoidance of oral participation; freezing when called upon
- Blushing, shaking, or voice tremor when speaking
- Avoidance of group tasks, presentations, or shared meals
- In adults: avoidance of networking, presentations, job interviews, and phone calls

SEREN Detection Signals:

- Gaze-aversion frequency during simulated social tasks
- Voice-tremor detection via acoustic jitter and shimmer analysis under simulated-audience conditions
- Response-delay differential between solo and group task conditions
- Avoidance-behaviour scoring on social-scenario tasks
- Facial micro-expression analysis for discomfort or fear during speaking prompts
- Camera-based heart-rate proxy during social tasks

Scientific Basis: *Clark & Wells (1995); Rapee & Heimberg (1997); Heimberg et al. (2010).*

Additional Teen-Presenting Conditions (Summary)

Condition	Category	Prevalence	Target Accuracy
Cluttering	Speech & Language	~0.5% global	82-88%
Panic Disorder	Anxiety Disorders	2-3% teen and above	78-84%
Adolescent Depression	Emotional & Behavioural	~8-11% teens	80-87%
Adjustment Disorder (Transition-Linked)	Emotional & Behavioural	~5-8% adolescents	73-80%
Gifted Underachievement	Silent Profiles & Neurodiversity	5-10% global	75-82%

Condition	Category	Prevalence	Target Accuracy
Body Image Insecurity	Insecurities & Confidence Deficits	~20% teens	73-79%
Relationship Insecurity	Insecurities & Confidence Deficits	~30% teen and above	72-78%
FOMO / Social Comparison Anxiety	Insecurities & Confidence Deficits	~40% teens	70-77%
Digital / Online Social Insecurity	Insecurities & Confidence Deficits	~35% teens	70-77%
Performance Anxiety (Sport/Arts/Music)	Anxiety Disorders	~15-25% competitive youth	76-83%

2.3 Adult Conditions (Ages 18-60+) — Persistence, Masking & Confidence

Adult presentations fall into two categories: childhood-onset conditions that persisted undiagnosed into working life, and conditions that develop specifically in adult contexts — career, workplace identity, and long-term relationship dynamics. SEREN's adult module is designed around short, functional, workplace-compatible sessions rather than remedial pedagogy.

Masking and Camouflaging

Category: Silent Profiles & Neurodiversity | **Global Prevalence:** ~5% global | **SEREN Target Accuracy:** 75-82%

Neurological / Psychological Basis: Active concealment of neurodivergent traits in order to appear neurotypical; cognitively and emotionally exhausting, and associated with burnout in adulthood; more commonly documented in girls and women.

Manifestations:

- Public performance appears typical while private struggle remains significant
- Marked exhaustion following social interaction or performance situations
- Inconsistent performance on identical tasks depending on context
- Preference for silent failure over visibly asking for help
- Heavy reliance on memorisation to compensate for an underlying decoding deficit

SEREN Detection Signals:

- Gap between task accuracy and effort-proxy indicators
- Sharp performance decline in the final third of a session (fatigue curve)
- Standard deviation of performance across repeated identical tasks
- Frequency of available hints or supports never used
- Comparison of memorisation accuracy against decoding accuracy

Scientific Basis: Hull et al. (2017); Cage & Troxell-Whitman (2019); Cook et al. (2021).

Additional Adult-Presenting Conditions (Summary)

Condition	Category	Prevalence	Target Accuracy
Health Anxiety (Illness Anxiety)	Anxiety Disorders	~4-5% adults	72-79%
Burnout Syndrome (Adult/Workplace)	Emotional & Behavioural	~20-30% workforce	74-81%
Intelligence / Imposter Syndrome	Insecurities & Confidence Deficits	~70% of professionals	75-81%
Academic Trauma (Fixed-Identity Damage)	Insecurities & Confidence Deficits	~20% adults (school-carried)	73-79%
Workplace / Career Insecurity	Insecurities & Confidence Deficits	~35% workforce	72-78%
Stuttering-Linked Confidence Deficit (Adult)	Insecurities & Confidence Deficits	~20% of adult stutterers	85-90%

Condition	Category	Prevalence	Target Accuracy
Decision Paralysis / Choice Anxiety	Insecurities & Confidence Deficits	~20% adults	70-77%
Financial Confidence Insecurity (Adult)	Insecurities & Confidence Deficits	~40% working adults (India)	68-75%
Leadership Avoidance Pattern (Adult)	Insecurities & Confidence Deficits	~30% workforce (esp. women)	70-77%
Time-Blindness (Temporal Processing Deficit)	Attention & Processing	Common in ADHD (70-80%); ~8% general	76-83%
Voice Disorder (Functional/Habitual Dysphonia)	Speech & Language	~6-9% general population	80-87%
Accent-Related Communication Anxiety (Adult)	Speech & Language	Not formally tracked; common in ESL professionals	70-77%
Undiagnosed Adult ADHD (Compensated/Masked)	Attention & Processing	35-78% of childhood ADHD persists	85-91%
Undiagnosed Adult Dyslexia (Compensated)	Learning Disorders	Lifelong persistence of childhood dyslexia	88-93%
Career-Stagnation Confidence Collapse	Insecurities & Confidence Deficits	Not formally tracked; workplace-linked	68-75%
Parental Burnout and Confidence Erosion	Emotional & Behavioural	~35-45% of parents (self-report)	70-77%

2.4 All-Ages / Cross-Cutting Conditions

A substantial group of conditions in the SEREN catalogue present meaningfully across the full lifespan — from generalised anxiety to word-finding difficulty to a wide spectrum of confidence and insecurity-based patterns. These are screened using age-adapted task variants of the same underlying signal architecture.

Generalised Anxiety Disorder (GAD)

Category: Anxiety Disorders | **Global Prevalence:** 6.5% children; 3.7% adults | **SEREN Target Accuracy:** 85-90%

Neurological / Psychological Basis: Associated with dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis and reduced prefrontal control over amygdala-driven worry circuitry.

Manifestations:

- Excessive worry spanning multiple domains: school, family, friendships, and the future
- Difficulty concentrating; mind going blank during tasks
- Physical complaints such as headaches or stomach aches before school
- Repeated reassurance-seeking from trusted adults
- Perfectionism, with excessive time spent on tasks and distress at imperfection
- Sleep difficulty, fatigue, and irritability

SEREN Detection Signals:

- Worry-language frequency detected via NLP analysis of task-related self-talk
- Perfectionism score derived from erase/redo action counts on touchscreen tasks
- Task-abandonment rate above a defined threshold, indicating failure-avoidance
- Response-latency elongation on low-difficulty items, indicating anxious overthinking
- Performance delta between timed and self-paced task versions

Scientific Basis: Behar et al. (2009); Borkovec & Ruscio (2001).

Additional All-Ages Conditions (Summary)

Condition	Category	Prevalence	Target Accuracy
Word-Finding Difficulty (Anomia)	Speech & Language	5-7% global	85-90%
Social Anxiety Disorder (SAD)	Anxiety Disorders	7-13% lifetime	87-92%
Exam / Test Anxiety	Anxiety Disorders	10-40% of students	85-90%
Emotional Dysregulation	Emotional & Behavioural	~8% global	82-87%
Trauma-Based Silence / Situational Mutism	Emotional & Behavioural	~5% global	75-82%
Silent Introversion-Driven Academic Suppression	Silent Profiles & Neurodiversity	~30-50% (personality trait)	75-82%
Masking and Camouflaging	Silent Profiles & Neurodiversity	~5% global	75-82%

Condition	Category	Prevalence	Target Accuracy
High-Functioning Autism (Masked)	Silent Profiles & Neurodiversity	1-2% global	80-86%
Invisible Struggle (Subclinical Composite Burden)	Silent Profiles & Neurodiversity	~15% global	70-77%
Alexithymia (Difficulty Identifying/Naming Emotion)	Silent Profiles & Neurodiversity	~10% general population	72-79%
Shyness (Deep-Rooted)	Insecurities & Confidence Deficits	~15% all ages	78-84%
Voice and Expression Insecurity	Insecurities & Confidence Deficits	~25% all ages	78-84%
Rejection Sensitivity (RSD)	Insecurities & Confidence Deficits	~30% in ADHD; ~15% general	78-84%
Social Belonging Insecurity	Insecurities & Confidence Deficits	~20% all ages	73-79%
Performance / Achievement Insecurity	Insecurities & Confidence Deficits	~25% all ages	75-81%
Anger as Hidden Insecurity	Insecurities & Confidence Deficits	~10% all ages	73-79%
Family-Origin Insecurities	Insecurities & Confidence Deficits	~30% all ages	70-76%
Perfectionism-Driven Self-Criticism	Insecurities & Confidence Deficits	~25-30% students and professionals	73-80%
Public Speaking Phobia (Glossophobia)	Insecurities & Confidence Deficits	~75% general population (some degree)	82-88%
Executive Function Deficit (Planning and Organisation)	Attention & Processing	~10% school and working population	80-87%
Grief-Related Withdrawal (Unprocessed Loss)	Emotional & Behavioural	Situational; ~5-7% at any time	70-77%
Highly Sensitive Person (HSP) Overwhelm Profile	Silent Profiles & Neurodiversity	~15-20% population	72-79%
Chronic People-Pleasing / Fawn Response	Silent Profiles & Neurodiversity	~15% general population	72-78%

SECTION 3 — Detection Technology: Complete App-Based Architecture for 95-99% Accuracy

SEREN's target accuracy of 95-99% for the highest-confidence condition categories is achieved through fusion across 12 independently validated detection modalities. No single modality reaches this accuracy alone. The architecture combines computer vision, handwriting analysis, speech and audio machine learning, natural language processing, cognitive task batteries, passive device sensors, camera-based physiological sensing, and structured teacher/parent input — all running on a standard Android mobile device, fully offline.

3.1 Technology Stack Overview — 12 Modalities

Modality	Technology Used	Conditions Covered	Accuracy Contribution
1. Computer Vision & Eye Tracking	MediaPipe Iris, OpenCV, TFLite	Dyslexia, ADHD, Social Anxiety, Depression, Autism, Stuttering	85-92% (dyslexia alone)
2. Handwriting & Drawing CNN	TensorFlow Lite CNN, stylus pressure API, stroke analysis	Dyslexia, Dysgraphia, DCD, Dyscalculia	up to 99%
3. Speech & Audio ML	On-device Whisper, custom disfluency model, Librosa	Stuttering, Cluttering, APD, Word-Finding, Selective Mutism	90-95%
4. NLP Signal Detection	On-device BERT-family model, custom sentiment model	All insecurities, Depression, Anxiety, GAD, RSD, Academic Trauma	80-88%
5. Reaction Time & Latency Engine	Nanosecond-precision Android timer, statistical analysis	Processing Speed, ADHD, APD, Dyslexia, Anxiety (overthinking)	88-93%
6. Passive Device Sensor Fusion	Android SensorManager: accelerometer, gyroscope, touch, pressure	ADHD Hyperactive, DCD, restlessness, sensory processing	78-85%
7. Cognitive Task Battery	Custom gamified CPT, Go/No-Go, N-back, Stroop, spatial rotation	ADHD, Processing Speed, Working Memory, Spatial, Executive Function	85-92%
8. Facial Emotion Recognition	MediaPipe Face Mesh (468 landmarks), on-device emotion model	Anxiety, Social Anxiety, Depression, Trauma, Selective Mutism	80-87%
9. Camera-Based Physiology (rPPG)	Remote photoplethysmography via front camera	Anxiety, Panic, Social Anxiety, RSD, Exam Anxiety, Stress	75-83%
10. Gamified Hidden Assessment	Custom game engine; hidden-task paradigms	All conditions — covert assessment without demand characteristics	+5-8% accuracy boost
11. Structured Input (Teacher/Parent/Self)	Digital questionnaires adapted from validated scales	All conditions — cross-validation against gold-standard measures	+8-12% accuracy boost
12. Longitudinal Analytics	On-device storage with encrypted sync; trend analysis	All conditions — baseline, follow-up, intervention response	+10-15% accuracy over time

3.2 Modality-by-Modality Technical Specification

Modality 1 — Computer Vision & Webcam Eye Tracking

Eye-tracking-based dyslexia research consistently reports that affected readers show longer fixation durations, higher regression rates, shorter saccade lengths, and atypical refixation patterns relative to typical readers. Recent narrative reviews report AI-based dyslexia detection from eye-movement data alone reaching above 80% accuracy; SEREN's multimodal fusion targets 92-95% by combining this signal with handwriting and speech data.

Component	What It Measures	Conditions Flagged
Iris landmark tracking	Gaze direction, fixation point, saccade trajectory	Dyslexia, ADHD, Social Anxiety, Autism
Blink-rate analysis	Blink frequency and duration under cognitive load	ADHD, Anxiety, Fatigue
Pupil dilation proxy	Approximate cognitive load and stress response	Anxiety, cognitive load (fused with rPPG)
Reading gaze analysis	Fixation duration, regression count, saccade amplitude on a reading passage	Dyslexia, Processing Speed
Gaze deviation during instruction	On-task vs. off-task gaze during instruction screens	ADHD Inattentive, Anxiety
Facial emotion recognition	Seven basic emotion classes in real time	Anxiety, Depression, Social Anxiety, Selective Mutism
Micro-expression detection	Brief involuntary expressions under 500ms via optical flow	Social Anxiety, Trauma, Stuttering anticipatory anxiety

Modality 2 — Handwriting & Drawing CNN Analysis

Handwriting-based machine learning is among the highest-accuracy modalities in the SEREN stack. Published work using kinematic, temporal, and spatial features from handwriting samples reports accuracy up to 99% for early screening of learning disabilities (Zaibi & Bezine, 2024), and comparable results have been reported for combined dyslexia-dysgraphia screening frameworks (Rangasrinivasan et al., 2025).

Feature Category	Features Extracted	Conditions Detected	Expected Accuracy
Kinematic features	Velocity, acceleration, jerk coefficient per stroke; inter-stroke pause	Dysgraphia, DCD, Dyslexia	95-99%
Spatial features	Letter height variability, spacing consistency, baseline drift	Dysgraphia, Dyslexia (reversals)	92-97%
Temporal features	Writing speed, inter-word pause, task completion time	Processing Speed, DCD, Dysgraphia	88-93%
Pressure features	Touch-area proxy for pen pressure and pressure variation	DCD, Dysgraphia, anxiety-related grip	85-92%
Letter-formation CNN	Stroke direction and reversal detection against formation templates	Dyslexia, Dysgraphia	95-99%

Modality 3 — Speech & Audio Machine Learning

Component	Key Signals	Conditions Detected	Accuracy
Speech recognition & transcription	On-device speech-to-text pipeline	Stuttering, Cluttering, Selective Mutism, Word-Finding	90-95%
Disfluency detection	Acoustic feature model (MFCC, spectral centroid, zero-crossing rate)	Stuttering, Cluttering	88-93%
Voice stress analysis	Jitter, shimmer, harmonics-to-noise ratio	Social Anxiety, Selective Mutism, Voice Insecurity	80-87%
Phonological processing	Phoneme-level accuracy and RAN latency	Dyslexia, APD, Expressive Language Disorder	88-93%
Auditory discrimination	Accuracy vs. stepped signal-to-noise ratio	APD, Dyslexia (phonological), Receptive Language	82-88%
RAN (Rapid Automatized Naming)	Naming speed, error rate, consistency	Dyslexia, Processing Speed, Word-Finding	88-93%
Speech rate & rhythm	Syllables per minute; inter-word pause distribution	Cluttering, Stuttering, Expressive Language	85-90%

Modality 4 — Natural Language Processing (NLP) Signal Detection

NLP analyses the content of verbal and text-based responses across all tasks. On-device language models process sentiment, worry language, comparison language, help-seeking versus refusal patterns, and fixed-identity statements. This modality carries primary signal weight for the insecurity and emotional condition clusters, where direct sensor measurement alone is insufficient.

NLP Signal	Conditions Flagged	Example Signal Pattern
Sentiment analysis	Depression, GAD, Academic Trauma, RSD	Frequency of negative self-referential statements
Worry-language detection	GAD, Separation Anxiety, School Refusal	Conditional and catastrophising phrase patterns
Comparison language	FOMO, Belonging Insecurity, Body Image	Comparative self-vs-other constructions
Fixed-identity language	Academic Trauma, Imposter Syndrome	Stable-attribute self-descriptions of inability
Help-refusal pattern	Intelligence Insecurity, Masking, 2e	Hint features never used; explicit help refusal
Apologetic language	Voice Insecurity, Social Anxiety, RSD	Pre-emptive qualifying and apologetic phrasing
Hopelessness markers	Depression, Anxiety, Academic Trauma	Future-negative constructions

Modality 5 — Reaction Time & Response Latency Engine

SEREN logs response latency with nanosecond precision on every task. The latency distribution — not merely the mean — carries diagnostic signal: bimodal distributions suggest ADHD-linked inconsistency; uniformly elevated latency suggests a processing-speed deficit; latency elongation specifically on verbal tasks suggests phonological retrieval difficulty; and latency elongation on easy items suggests anxious overthinking. All latency scores are z-scored against age-stratified normative data collected during pilot deployment.

Modality 6 — Passive Device Sensor Fusion

Sensor	Signal Extracted	Conditions Flagged
Accelerometer	Movement frequency, fidgeting intensity, restlessness index	ADHD Hyperactive, DCD, Sensory Processing
Gyroscope	Rotation rate, postural instability, balance proxy	DCD, Dyspraxia, Sensory Processing
Touch pressure	Grip intensity, pressure variability, stress-grip proxy	Anxiety, Dysgraphia, DCD
Touch area (size)	Contact-surface area as a pen-pressure proxy	Dysgraphia, stylus-based writing tasks
Screen interaction	Tap frequency, scroll speed, swipe irregularity, session interruptions	ADHD, restlessness, disengagement
App lifecycle events	Time-on-task, background exits, re-entry latency	ADHD, School Refusal, avoidance patterns

Modality 7 — Cognitive Task Battery

Task	Paradigm	Conditions Validated	Accuracy
Continuous Performance Test (CPT)	Sustained multi-minute attention task	ADHD Inattentive, Processing Speed	88-93%
Go/No-Go	Respond to Go stimulus; withhold on No-Go	ADHD Impulsive, Anxiety	85-92%
N-Back (1-back / 2-back)	Working-memory match task, visual and auditory	ADHD, Processing Speed, Cognitive Load	82-88%
Stroop Task	Incongruent colour-word naming	ADHD, Executive Function, Anxiety	85-90%
Digit Span (forward/backward)	Sequence recall, audio and visual versions	Working Memory (ADHD, Dyslexia)	80-87%
Spatial Rotation Task	Match rotated 3D object to target	Spatial Intelligence, DCD, Giftedness	78-85%
Simplified Matrix Reasoning	Visual pattern completion, forced choice	Non-verbal ability, Giftedness, 2e	80-87%
Subitizing / Dot Enumeration	Rapid dot-count task	Dyscalculia, Number Sense	85-90%
Tower of London (simplified)	Move objects to match target in fewest moves	Executive Function, ADHD, Planning	78-84%
Emotional Stroop	Colour-naming of emotional vs. neutral words	Anxiety, Trauma, Social Anxiety	78-85%

Modality 8 & 9 — Facial Emotion Recognition & Camera-Based Physiological Sensing (rPPG)

Facial emotion recognition uses landmark tracking across roughly 468 facial points combined with an on-device emotion classification model spanning seven basic emotion categories, plus action-unit analysis for micro-expressions. It is used to detect fear or anxiety during social tasks, emotional shutdown during trauma-linked silence, anticipatory anxiety before stuttering-trigger sounds, and flat affect associated with depression.

Remote photoplethysmography (rPPG) estimates heart rate and heart-rate variability from subtle colour changes in facial skin captured by the front camera, using bandpass-filtered signal processing.

Published studies report 80-90% correlation with pulse-oximetry-derived heart rate under controlled conditions. rPPG is used as a supplementary physiological-arousal signal — particularly for anxiety, panic, and stress-response quantification — and is weighted lower under motion or variable lighting, where its accuracy degrades.

Modality 10 — Gamified Hidden Assessment Tasks

Hidden assessment tasks embed validated cognitive and behavioural measures inside engaging game formats, removing demand characteristics — the tendency for a child or adult to perform differently once they know they are being formally tested. This is particularly valuable for anxious users who underperform on explicit tests, for masking individuals who consciously control their presentation when aware of assessment, and for young children who cannot sustain motivation through a formal battery.

Game	Hidden Measure	Conditions Detected
Sort the Stars	Categorisation speed, cognitive flexibility, switch cost	ADHD, Executive Function, 2e
Mirror the Pattern	Visual-spatial working memory, pattern reproduction	Dyspraxia, Dyscalculia, Spatial ability
Catch the Sound	Auditory discrimination, sustained auditory attention	APD, ADHD (auditory)
Word Racer	Typing fluency, phonological encoding, spelling automaticity	Dyslexia, Dysgraphia
Memory Castle	Spatial working memory (Corsi-block analogue)	ADHD, Working Memory, Processing Speed
Tell Me a Story	Verbal elaboration, expressive language, sentiment content	Expressive Language, Anxiety, Depression
Brave Explorer	Approach vs. avoidance decision pattern	Social Anxiety, Selective Mutism, Shyness
Emotion Town	Emotion recognition accuracy, social problem-solving	Autism, Pragmatic Language, Social Communication

3.3 Complete Condition x Technology Matrix

Every condition in the 80+ catalogue is mapped in the SEREN technical appendix to its specific combination of primary modalities, exact in-app task features, and projected accuracy. A representative excerpt is shown below; the complete matrix spans all 80+ conditions.

Condition	Primary Modalities	Key App Features	Target Accuracy
Dyslexia	Eye tracking + Handwriting CNN + Speech ML	Letter copying, reading-gaze task, RAN timing	95-99%
ADHD (all types)	CPT + Go/No-Go + Gaze + Sensors + Longitudinal	Sustained attention task, gaze deviation, session exits	90-96%
Stuttering	Speech ML + Facial + Longitudinal	Disfluency detection, secondary-behaviour video	90-95%
Social Anxiety	Facial + rPPG + Speech ML + Reaction time	Gaze avoidance, voice tremor, heart-rate proxy	87-92%

Condition	Primary Modalities	Key App Features	Target Accuracy
Sensory Processing Disorder	Passive sensors + Performance delta + Cognitive	Noise sensitivity, pressure patterns	78-84%
Imposter Syndrome	NLP + Reaction time + Cognitive tasks	Help refusal, answer hiding, success minimisation	75-81%

3.4 Multimodal AI Fusion Architecture — Achieving 95-99% Accuracy

No single modality achieves 95%+ accuracy alone. SEREN's core architectural innovation is the fusion of up to 12 independently validated modalities into a single composite risk score per condition, using a weighted ensemble approach trained on pilot deployment data.

Modalities Active	Expected Accuracy	Example Condition	Fusion Method
1 modality only	65-75%	Dyslexia via eye tracking alone	Single-model output
2-3 modalities	78-85%	Dyslexia: eye tracking + handwriting + phonological	Weighted average; highest-confidence model leads
4-6 modalities	87-93%	ADHD: CPT + gaze + sensors + Go/No-Go + teacher input	Random-forest ensemble; feature-importance weighting
7-9 modalities	93-96%	Anxiety: facial + rPPG + NLP + latency + abandonment	Gradient-boosted tree on fused feature vector
10-12 modalities + longitudinal	96-99%	Full SEREN session with longitudinal baseline comparison	Deep fusion model; on-device inference with uncertainty quantification
+ Teacher + Parent input	+8-12% accuracy boost across all conditions	Any condition	Structured evidence cross-validation; Bayesian update

3.5 Libraries, Frameworks & On-Device ML Stack

Library / Framework	Purpose	Why On-Device
TensorFlow Lite	All CNN and ML inference: handwriting, emotion, NLP, gaze	Offline-first; low-latency inference on mid-range Android hardware
MediaPipe	Face Mesh, Iris tracking, hand landmarks, pose estimation	Mobile-optimised; real-time performance on entry-level devices
OpenCV (Android)	Image processing, optical flow, rPPG signal extraction	Native image-processing performance
On-device speech model (Whisper-class, compact)	Speech recognition, phoneme analysis, disfluency detection	Runs fully offline; supports Hindi, Marathi, and other Indian languages
Audio feature preprocessing (Librosa-class)	MFCC, spectral, jitter, shimmer extraction at training time	Enables compact on-device audio inference models
On-device NLP toolkit	Sentiment, entity, language identification, offline speech backup	Works fully offline after initial model load

Library / Framework	Purpose	Why On-Device
Encrypted on-device data store	Session data, longitudinal tracking, progress history	All data remains on-device by default; encrypted at rest
Android SensorManager API	Accelerometer, gyroscope, pressure — passive sensor collection	Native API; negligible battery overhead
CameraX	Front-camera management for eye tracking, rPPG, facial analysis	Lifecycle-aware; automatic permission handling
Background sync scheduler	Encrypted, anonymised sync of aggregate data when connectivity is available	Battery-aware; privacy-preserving by design
Cross-platform game engine (optional layer)	Gamified hidden-assessment tasks with rich interaction	Enables richer engagement for younger users

SECTION 4 — Intervention Architecture: Condition-by-Condition Treatment in App

4.1 Evidence-Based Intervention Principles

SEREN's intervention layer rests on three principles. First, structure, cumulation, and explicitness: interventions follow logical, cumulative sequences engaging multiple sensory modalities, consistent with structured literacy approaches such as Orton-Gillingham. Second, gamification meaningfully improves outcomes: systematic reviews of serious games report significant literacy gains, and gamified phonics instruction has been shown to outperform traditional delivery in comparable studies. Third, personalisation and adaptive difficulty: an on-device model continuously adjusts task difficulty to maintain an approximate 80% success rate — the empirically supported “flow state” zone that avoids both boredom and anxiety.

4.2 Intervention Catalogue by Category

Every intervention runs in short daily sessions (typically 10-20 minutes), fully offline, with adaptive difficulty and age-appropriate framing from age 5 through 60+. The table below summarises representative interventions across the major condition categories; the complete condition-by-condition intervention catalogue for all 80+ conditions is maintained in the SEREN product specification.

Condition	Representative Interventions	Evidence Base
Dyslexia	Phoneme segmentation games, spaced-repetition sight-word drills, syllable and word-family building	Orton-Gillingham; National Reading Panel (2000); Ehri (2005)
Dyscalculia	Subitizing games, number-line navigation, spaced-repetition fact retrieval, place-value trading tasks	Dehaene (2011); Butterworth; Fuchs (2005)
Dysgraphia	Guided letter-formation tracing, air-writing via camera, graphic-organiser story planning	Graham & Harris; Berninger (2006)
DCD / Dyspraxia	Motor-sequence builders, hand-eye coordination challenges, daily-routine sequencing practice	Polatajko & Cantin (2006); Henderson
ADHD	Sustained-attention games, inhibition-training tasks, working-memory challenges, breathing-based self-regulation	Klingberg (2005); Barkley (2015); Zoogman (2015)
Auditory Processing Disorder	Figure-ground listening training, dichotic-listening games, temporal-processing drills	Bellis (2003); Chermak & Musiek
Condition	Representative Interventions	Evidence Base
Stuttering	Easy-onset practice, fluency-shaping coaching, voluntary-stutter desensitisation, confidence-building speaking challenges	Guitar (2014); Bricker-Katz (2009)
Word-Finding Difficulty	Retrieval-cue training, semantic-web building, category-fluency drills	Nickels (2002); Levelt (1989)
Generalised Anxiety Disorder	Worry externalisation, structured breathing, CBT-based thought restructuring, graduated exposure	Beck (1979); Borkovec (2001)
Social Anxiety	Graduated exposure hierarchy, decatastrophising practice, guided speaking-confidence exercises	Clark & Wells (1995); Rapee & Heimberg (1997)

Condition	Representative Interventions	Evidence Base
Selective Mutism	Stimulus-fading from whisper to full voice, graduated social-distance practice	McHolm (2005); Kehle (1998)
Childhood Depression (masked)	Behavioural activation planning, positive-moment logging, daily mood check-in	Lewinsohn behavioural activation; Kovacs (1985)
Emotional Dysregulation	Trigger mapping, DBT-informed distress-tolerance skills, structured calm-down routines	Linehan (1993) DBT; Gross (2002)
Condition	Representative Interventions	Evidence Base
Shyness / Social Confidence	Graduated micro-courage challenges, strength identification, voice-volume training	Zimbardo (1977); positive psychology literature
Body Image Insecurity	Body-neutrality exercises, media-literacy training, values-based identity building	Cash (2008); Tylka (2011)
Imposter Syndrome	Evidence-based success logging, reattribution training, competence-portfolio building	Clance & Imes (1978); Dweck (2006)
Rejection Sensitivity	Pause-and-check protocols, cognitive reappraisal, graduated exposure to mild criticism	Dodson (2016); DBT interpersonal effectiveness
Voice & Expression Insecurity	Vocal-projection coaching, assertive-sentence practice, camera-based posture feedback	Alberti & Emmons (2008); Rakos (1991)

4.3 Adult-Specific Intervention Protocols

Adults who discover childhood-onset conditions through SEREN receive a structurally different protocol from children. Four principles govern this design: (1) psychoeducation comes first — reframing the condition as a specific, nameable neurological or psychological pattern rather than a personal failing is consistently the single most impactful early intervention for adults; (2) interventions are functional rather than remedial — an adult with dyslexia typically does not need phonics instruction, but does need compensatory strategies suited to professional contexts; (3) sessions are short and workplace-compatible, typically 15 minutes, deployable during a commute or a lunch break; and (4) outcomes are measured functionally (task completion, confidence, avoidance reduction) rather than against paediatric literacy benchmarks. A 2024 study confirmed that 24 sessions of 15-20 minutes each produced measurable literacy improvement in adults aged 16-30 with dyslexia — SEREN's adult module is directly modelled on this evidence base, adapted for Indian languages and professional contexts.

4.4 Gamification Framework

Mechanic	SEREN Implementation	Why It Works
Adaptive Difficulty (Flow State)	On-device model adjusts task difficulty in real time to maintain ~80% success rate	Csikszentmihalyi (1990) flow theory
Progress Visualisation	Multi-week journey map with condition-specific milestones	Locke & Latham (2002) goal-setting theory
Immediate, Non-Punitive Feedback	Every response receives encouraging feedback or gentle redirection, never punitive correction	Bandura self-efficacy theory
Narrative Embedding	Activities framed within engaging story contexts	Green (2004) narrative transportation research
Mastery Badges	Condition-specific achievement identities (e.g. a reading-focused badge track)	Dweck (2006) growth mindset

Mechanic	SEREN Implementation	Why It Works
Family Celebration Mode	Parent-facing app surfaces milestone unlocks for shared celebration	Epstein (2001) parental-involvement research
Offline Streaks	Daily practice streaks with gentle, non-punitive reminders	Lally et al. (2010) habit-formation research
Hidden Assessment	Gamified tasks simultaneously assess and intervene	Reduces test anxiety; captures authentic performance

SECTION 5 — Accuracy Projections, Validation & Scientific Basis

5.1 Accuracy Projection by Condition Category

Condition Category	Modalities Used	Projected Accuracy	Primary Scientific Basis
Learning Disorders (Dyslexia, Dysgraphia)	Eye tracking + handwriting CNN + speech ML + longitudinal	95-99%	Zaibi & Bezine (2024); Sedmidubsky et al. (2024); Vajs et al. (2023)
Dyscalculia, DCD	Cognitive tasks + sensors + motor CNN + reaction time	88-93%	Butterworth (2010); Henderson MABC-2; Wilson et al. (2006)
ADHD (all types)	CPT + Go/No-Go + gaze + sensors + longitudinal + teacher input	92-96%	Conners (2000); Barkley (2015); Klingberg (2005)
Processing Speed, APD	Reaction time + audio ML + cognitive + longitudinal	88-93%	Shanahan et al. (2006); Bellis (2003)
Stuttering, Cluttering	Speech ML + facial + rPPG + longitudinal	90-95%	Guitar (2014); Chang et al. (2008)
Other Speech/Language Disorders	Speech ML + NLP + cognitive tasks + teacher input	82-90%	Leonard (1998); Bishop (1997); Levelt (1989)
Anxiety Disorders	Facial + rPPG + NLP + reaction time + teacher + longitudinal	85-93%	Clark & Wells (1995); Hembree (1988)
Emotional / Behavioural	NLP + facial + rPPG + cognitive + parent/teacher input	80-88%	Kovacs (1985); Linehan (1993)
Silent Profiles (Masking, 2e, Introversion)	Multi-modality inconsistency analysis + longitudinal	75-83%	Hull et al. (2017); Baum & Owen (2004); Cain (2012)
Insecurities & Confidence Deficits	NLP + facial + rPPG + gamified tasks + self-report + longitudinal	73-84%	Clance & Imes (1978); Dodson (2016); Cash (2008)

5.2 Interpreting SEREN's Accuracy Figures

SEREN is a screening tool, not a diagnostic tool. The accuracy figures throughout this paper represent the estimated probability that a person flagged by SEREN as at-risk for a given condition would be confirmed on formal clinical assessment (positive predictive value), or correctly cleared where SEREN does not flag risk (negative predictive value). The system is deliberately calibrated to prioritise sensitivity — catching genuine cases — over specificity, since the cost of a missed case is materially higher than the cost of an unnecessary professional referral. A positive SEREN screening result is designed to make that referral clearly and consistently warranted, reducing the burden on clinical capacity while substantially reducing the number of children and adults who go undetected.

5.3 Validation Roadmap

- Phase 1 — Instrument calibration: build age-stratified normative datasets for every latency, motor, and speech signal using pilot deployment cohorts.

- Phase 2 — Concurrent validity study: compare SEREN screening outputs against formal clinician assessment (SLD assessment protocols, ADHD rating scales, standardised anxiety and depression instruments) on a matched sample.
- Phase 3 — Longitudinal outcome study: track intervention-arm vs. control-arm outcomes over one academic year to validate that early flagging plus intervention improves measurable literacy, attention, and confidence outcomes.
- Phase 4 — External peer-reviewed publication of validation results, submitted to an appropriate educational-technology or child-development journal.

SECTION 6 — Product, Engineering & Go-To-Market Blueprint

This section sets out the concrete plan for taking SEREN from research specification to a launch-ready product: system architecture, data privacy and compliance posture, engineering roadmap, deployment model, policy alignment, business model, risk register, and launch readiness checklist.

6.1 System Architecture Overview

SEREN is architected as an offline-first Android application with an optional encrypted cloud-sync layer, structured across four logical layers:

Layer	Components	Design Notes
Presentation Layer	Native Android UI (Kotlin/Jetpack Compose) plus an optional cross-platform game engine layer for gamified hidden-assessment tasks	Age-adaptive UI (5-12 / 13-19 / 18-60+); multi-language support (English, Hindi, Marathi, and extensible to additional Indian languages)
Sensing & Signal Layer	On-device camera, microphone, touchscreen, and motion sensor pipelines feeding the 12 detection modalities	All raw sensor processing happens on-device; raw video/audio is never transmitted off-device by default
Inference Layer	TensorFlow Lite / MediaPipe models for CV, handwriting CNN, speech ML, NLP, and the multimodal fusion ensemble	All inference runs locally; model updates are delivered via periodic app updates rather than live cloud inference
Data & Sync Layer	Encrypted on-device relational store for session data and longitudinal history; optional background sync of anonymised, aggregated scores when connectivity is available	Sync is opt-in, encrypted in transit and at rest, and excludes raw biometric data

Companion Surfaces

- Teacher Dashboard — classroom-level view of screening status, flagged conditions, and recommended next steps, generated in plain, non-clinical language.
- Parent App — simplified summary of screening results, daily practice streaks, and milestone celebrations, available in the family's preferred language.
- Clinician / Referral Portal — structured export of screening data formatted for handoff to a psychologist, speech-language therapist, or special educator, explicitly marked as screening data requiring clinical confirmation.
- Adult Self-Screening Mode — a standalone entry point that does not require institutional (school) onboarding, designed for direct adult use.

6.2 Data Privacy, Security & Compliance

Given that SEREN processes sensitive biometric, behavioural, and child data, privacy-by-design is treated as a launch-blocking requirement rather than a post-launch feature.

- Offline-first default: all sensitive inference (handwriting, speech, facial, gaze) runs on-device; no raw video, audio, or handwriting image is transmitted to any server by default.
- Data minimisation: only derived numerical signals and risk scores — never raw biometric recordings — are eligible for optional cloud sync.

- **Consent architecture:** explicit parental consent is required before any child profile is created; explicit self-consent is required for adult users; consent is re-confirmed before any data leaves the device.
- **Encryption:** on-device data is encrypted at rest; any sync traffic is encrypted in transit using standard TLS.
- **Compliance alignment:** architecture is designed to align with India's Digital Personal Data Protection (DPDP) Act requirements for processing of children's personal data, including purpose limitation and parental consent, and is built to accommodate future compliance requirements (such as CBSE / state education department data-handling guidelines) as they are formalised.
- **No behavioural advertising:** SEREN does not monetise or share user data with third parties for advertising or unrelated commercial purposes under any circumstance.

6.3 Engineering Roadmap & Build Phases

Phase	Scope	Exit Criteria
Phase 0 — Foundation (Months 1-3)	Core Android app shell; on-device data store; consent and onboarding flows; integration of MediaPipe, TFLite, and the on-device speech pipeline	Working app skeleton with 3-4 flagship condition screens (Dyslexia, ADHD, Stuttering, Social Anxiety) functional end-to-end
Phase 1 — Core Modality Build-Out (Months 3-7)	Build out Modalities 1-7 (vision, handwriting, speech, NLP, latency engine, sensors, cognitive battery) across the full 80+ condition catalogue	All 12 modalities implemented; condition x technology matrix fully wired to live inference
Phase 2 — Fusion & Calibration (Months 6-9)	Train and calibrate the multimodal fusion ensemble; build age-stratified normative databases from pilot cohort data	Fusion model live; per-condition accuracy tracked against pilot validation data
Phase 3 — Intervention Layer (Months 7-11, overlapping)	Build the full gamified intervention catalogue mapped to Section 4; adaptive-difficulty engine; teacher/parent dashboards	Closed-loop screen-then-intervene experience live for all flagged conditions
Phase 4 — Pilot Deployment (Months 9-12)	Deploy in a small number of partner schools across the founding region plus a standalone adult self-screening release	Real-world usage data; first concurrent-validity comparisons against clinician assessment
Phase 5 — Scale-Readiness (Months 12-18)	Harden multi-language support, offline sync reliability, and low-end device performance; formal validation study; compliance audit	Launch-ready release candidate; validation study results ready for external publication

6.4 Deployment Model & Go-To-Market Plan

Deployment Channels

- **School-Deployed Channel** — SEREN deployed directly in classrooms as a Grade 1-8 screening layer, run by teachers with minimal training, aligned to school academic calendars and FLN assessment windows.
- **Direct-to-Parent Channel** — a consumer app entry point for parents to screen their child independently of school participation.
- **Direct-to-Adult Channel** — a standalone adult self-screening and intervention product, marketed independently of the school-facing product.

- Institutional / NGO Channel — partnerships with education-focused NGOs and state education departments for large-scale, subsidised deployment in underserved regions.

Go-To-Market Sequencing

- Stage 1 — Founding-region pilot: deploy in a focused set of partner schools to generate real validation data and refine the product before wider release.
- Stage 2 — State-level expansion: use pilot validation data to engage state education departments aligned with NIPUN Bharat implementation, positioning SEREN as an individual-level screening layer complementing existing FLN assessment tools.
- Stage 3 — National school-channel scale-up: expand school deployment nationally through a combination of direct sales to private schools and subsidised or free deployment in government schools via institutional and NGO partnerships.
- Stage 4 — Adult consumer scale-up: grow the direct-to-adult product independently once the underlying detection and intervention engine has been validated through the school channel.

6.5 NEP 2020 Policy Alignment & Institutional Fit

NEP 2020 designates universal Foundational Literacy and Numeracy (FLN) as the highest immediate priority of the Indian school system and established the National Mission on Foundational Literacy and Numeracy, operationalised through the NIPUN Bharat mission, targeting universal FLN by the end of Grade 3. The policy simultaneously commits to equitable and inclusive education, including identification and classroom-level support for children with learning difficulties, and calls for a pupil-teacher ratio structured to make individualised attention feasible.

SEREN is designed to sit directly inside this policy framework rather than alongside it. Where population-level FLN assessment tools measure aggregate literacy and numeracy attainment at the cohort level, SEREN provides the individual-level, condition-specific diagnostic layer that determines why a particular child has not yet reached foundational benchmarks — whether that is dyslexia, dyscalculia, attention difficulty, an anxiety-driven barrier, or a combination. This makes SEREN a natural technology partner for state-level NIPUN Bharat implementation teams, teacher-training programmes under NEP 2020's continuous professional development mandate, and Special Education Zones established under the policy for educationally disadvantaged regions.

- Direct fit with NIPUN Bharat's Grade 1-3 FLN identification requirement.
- Direct fit with NEP 2020's equitable and inclusive education mandate for identifying and supporting learning difficulties within mainstream classrooms.
- Regional-language, offline-first design matches NEP 2020's emphasis on mother-tongue-based foundational instruction.
- Zero-cost, teacher-administered model matches the resource constraints of Special Education Zones and socio-economically disadvantaged regions targeted by the policy.

6.6 Business Model & Sustainability

Revenue Stream	Model	Role in Business
Government / Institutional Licensing	Per-student or per-district licensing for state education departments and government school networks, positioned against FLN mission budgets	Primary long-term revenue and scale channel
Private School Direct Sales	Annual per-student or per-school subscription for private and international schools	Near-term revenue channel funding early product development
NGO / CSR Partnerships	Subsidised or grant-funded deployment in underserved government schools	Impact-driven channel supporting mission-aligned scale and validation data
Adult Consumer Subscription	Freemium screening with a paid subscription tier for the full adult intervention library	Independent consumer revenue stream, lower cost-to-serve
Clinician / Referral Network Partnerships	Referral-fee-free structured handoff partnerships with psychologists and speech-language therapists	Trust-building channel; not a primary revenue source

6.7 Risk Register & Mitigation

Risk	Severity	Mitigation
Screening false positives causing parental anxiety	Medium	Clear framing that results are screening, not diagnosis; every flagged result includes a plain-language explanation and a recommended next step, never a clinical label.
Screening false negatives (missed cases)	Medium-High	Deliberately calibrate for high sensitivity; combine with structured teacher/parent input as a cross-validation layer; recommend periodic re-screening rather than a single one-time assessment.
Data privacy or consent failure	High impact / Low likelihood if mitigated	Offline-first architecture; no raw biometric data leaves the device by default; explicit consent gating; independent privacy/compliance audit before wider launch.
Low-end device performance limitations	Medium	Maintain a lightweight inference profile for low-cost Android devices; degrade gracefully to lower-modality-count sessions rather than failing outright.
Teacher adoption friction	Medium	Minimal-training onboarding flow; teacher dashboard designed around existing classroom routines rather than requiring new clinical workflows.
Regulatory classification as a medical device	Medium	Maintain explicit screening (not diagnostic) positioning in all product and marketing language; engage relevant regulatory guidance early in the validation phase.
Validation study delays	Medium	Begin pilot data collection from Phase 0 rather than waiting for a dedicated validation phase, so real-world data accumulates continuously.

6.8 Launch Readiness Checklist

- Core detection engine covering at minimum the highest-prevalence flagship conditions (Dyslexia, Dyscalculia, Dysgraphia, ADHD, Stuttering, Social Anxiety, GAD) is fully functional end-to-end on-device.
- Multimodal fusion model calibrated against an initial normative dataset from pilot deployment.
- Teacher dashboard and parent app both functional, translated, and tested with real users.

- Consent flows, data encryption, and offline-first data handling independently reviewed for compliance.
- Intervention library live for every flagship condition included in the initial detection release.
- Pilot deployment completed in at least one partner-school cohort with documented outcome data.
- Adult self-screening entry point live and functionally independent of the school-onboarding flow.
- Risk register reviewed and mitigations implemented for all high-severity items prior to public launch.

References — Complete Scientific Bibliography

- Adams, M.J. (1990). *Beginning to Read: Thinking and Learning about Print*. MIT Press.
- Alberti, R.E., & Emmons, M.L. (2008). *Your Perfect Right: Assertiveness and Equality in Your Life and Relationships*. Impact Publishers.
- APA (2013). *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)*. American Psychiatric Association.
- Aron, E.N. (1996). *The Highly Sensitive Person*. Broadway Books.
- ASHA (2005). *(Central) Auditory Processing Disorders*. American Speech-Language-Hearing Association Technical Report.
- Attwood, T. (2007). *The Complete Guide to Asperger's Syndrome*. Jessica Kingsley Publishers.
- Ayres, A.J. (1972). *Sensory Integration and Learning Disorders*. Western Psychological Services.
- Barkley, R.A. (2015). *Attention-Deficit Hyperactivity Disorder: A Handbook for Diagnosis and Treatment (4th ed.)*. Guilford Press.
- Baron-Cohen, S. (2000). Theory of Mind and Autism: A Review. *International Review of Research in Mental Retardation*, 23, 169-184.
- Baum, S.M., & Owen, S.V. (2004). *To Be Gifted and Learning Disabled*. Creative Learning Press.
- Baumeister, R.F., & Leary, M.R. (1995). The Need to Belong: Desire for Interpersonal Attachments as a Fundamental Human Motivation. *Psychological Bulletin*, 117(3), 497-529.
- Beck, A.T. (1979). *Cognitive Therapy of Depression*. Guilford Press.
- Behar, E., DiMarco, I.D., Hekler, E.B., Mohlman, J., & Staples, A.M. (2009). Current Theoretical Models of Generalized Anxiety Disorder. *Journal of Anxiety Disorders*, 23(8), 1011-1023.
- Bellis, T.J. (2003). *Assessment and Management of Central Auditory Processing Disorders in the Educational Setting*. Delmar Learning.
- Berninger, V.W. (2006). Research-Supported Ideas for Implementing Reauthorized IDEA with Intelligent Professional Judgment. *Learning Disabilities Research & Practice*, 21(3).
- Bishop, D.V.M. (1997). *Uncommon Understanding: Development and Disorders of Language Comprehension in Children*. Psychology Press.
- Bishop, D.V.M., & Norbury, C.F. (2002). Exploring the Borderlands of Autistic Disorder and Specific Language Impairment. *Journal of Child Psychology and Psychiatry*, 43(7).
- Bloodstein, O., & Ratner, N.B. (2008). *A Handbook on Stuttering (6th ed.)*. Delmar Cengage Learning.
- Borkovec, T.D., & Ruscio, A.M. (2001). Psychotherapy for Generalized Anxiety Disorder. *Journal of Clinical Psychiatry*, 62(Suppl 11).
- Bowlby, J. (1988). *A Secure Base: Parent-Child Attachment and Healthy Human Development*. Basic Books.
- Bricker-Katz, G., Lincoln, M., & McCabe, P. (2009). A Life-Time of Stuttering: How Emotional Reactions to Stuttering Shape Attitudes and Perceptions of Work Performance. *Disability and Rehabilitation*, 31(23).
- Butterworth, B. (2010). Foundational Numerical Capacities and the Origins of Dyscalculia. *Trends in Cognitive Sciences*, 14(12), 534-541.
- Cage, E., & Troxell-Whitman, Z. (2019). Understanding the Reasons, Contexts and Costs of Camouflaging for Autistic Adults. *Journal of Autism and Developmental Disorders*, 49(5).
- Cain, S. (2012). *Quiet: The Power of Introverts in a World That Can't Stop Talking*. Crown Publishers.

- Cash, T.F. (2008). *The Body Image Workbook: An Eight-Step Program for Learning to Like Your Looks*. New Harbinger Publications.
- Chang, S.E., Kenney, M.K., Loucks, T.M.J., & Ludlow, C.L. (2008). Brain Areas Underlying Visual Feedback Control of Speech Production in Normal-Speaking Adults. *NeuroImage*, 41(1).
- Chermak, G.D., & Musiek, F.E. (2007). *Handbook of (Central) Auditory Processing Disorder*. Plural Publishing.
- Christie, P., Duncan, M., Fidler, R., & Healy, Z. (2012). *Understanding PDA in Children*. Jessica Kingsley Publishers.
- Clance, P.R., & Imes, S.A. (1978). The Imposter Phenomenon in High Achieving Women. *Psychotherapy: Theory, Research & Practice*, 15(3).
- Clark, D.M. (1986). A Cognitive Approach to Panic. *Behaviour Research and Therapy*, 24(4).
- Clark, D.M., & Wells, A. (1995). A Cognitive Model of Social Phobia. In R. Heimberg (Ed.), *Social Phobia: Diagnosis, Assessment, and Treatment*. Guilford Press.
- Cohan, S.L., Chavira, D.A., & Stein, M.B. (2008). Practitioner Review: Psychosocial Interventions for Children with Selective Mutism. *Journal of Child Psychology and Psychiatry*, 47(3).
- Conners, C.K. (2000). *Conners' Rating Scales-Revised*. Multi-Health Systems.
- Cook, J., Hull, L., Crane, L., & Mandy, W. (2021). Camouflaging in Autism: A Systematic Review. *Clinical Psychology Review*, 89.
- Csikszentmihalyi, M. (1990). *Flow: The Psychology of Optimal Experience*. Harper & Row.
- Daly, D. (1996). *The Source for Stuttering and Cluttering*. LinguiSystems.
- Dehaene, S. (2011). *The Number Sense: How the Mind Creates Mathematics*. Oxford University Press.
- Deffenbacher, J.L. (1999). Cognitive-Behavioral Conceptualization and Treatment of Anger. *Journal of Clinical Psychology*, 55(3).
- Dodson, W. (2016). Rejection Sensitive Dysphoria and ADHD. *ADDitude Magazine*.
- Dweck, C.S. (2006). *Mindset: The New Psychology of Success*. Random House.
- Ehri, L.C. (2005). Learning to Read Words: Theory, Findings, and Issues. *Scientific Studies of Reading*, 9(2).
- Elliot, A.J., & Church, M.A. (1997). A Hierarchical Model of Approach and Avoidance Achievement Motivation. *Journal of Personality and Social Psychology*, 72(1).
- Fayyad, J. et al. (2007). Cross-National Prevalence and Correlates of Adult ADHD. *British Journal of Psychiatry*, 190.
- Frontiers in Human Neuroscience (2025). *Emerging Technologies and Neuroscience-Based Approaches in Dyslexia: A Narrative Review*.
- Gathercole, S.E., & Alloway, T.P. (2008). *Working Memory and Learning: A Practical Guide for Teachers*. SAGE Publications.
- Google (2024). *MediaPipe Face Mesh and Iris Landmark Model Documentation*.
- Gross, J.J. (2002). Emotion Regulation: Affective, Cognitive, and Social Consequences. *Psychophysiology*, 39(3).
- Guitar, B. (2014). *Stuttering: An Integrated Approach to Its Nature and Treatment* (4th ed.). Lippincott Williams & Wilkins.
- Harmer, C.J. et al. (2011). Why Do Antidepressants Take So Long to Work? *British Journal of Psychiatry*.
- Hazan, C., & Shaver, P. (1987). Romantic Love Conceptualized as an Attachment Process. *Journal of Personality and Social Psychology*, 52(3).
- Hembree, R. (1988). Correlates, Causes, Effects, and Treatment of Test Anxiety. *Review of Educational Research*, 58(1).
- Henderson, S.E., & Sugden, D.A. *Movement Assessment Battery for Children-2 (MABC-2)*. Pearson Assessment.
- Herman, J.L. (1992). *Trauma and Recovery*. Basic Books.

- Hewlett, S.A. (2014). *Executive Presence*. HarperBusiness.
- Hull, L., Petrides, K.V., Allison, C., Smith, P., Baron-Cohen, S., Lai, M.C., & Mandy, W. (2017). 'Putting on My Best Normal': Social Camouflaging in Adults with Autism Spectrum Conditions. *Journal of Autism and Developmental Disorders*, 47(8).
- Immordino-Yang, M.H. (2016). *Emotions, Learning, and the Brain*. W.W. Norton.
- Johnson, S.M. (2008). *Hold Me Tight: Seven Conversations for a Lifetime of Love*. Little, Brown.
- Joseph, A., & Devu, B.K. (2022). Prevalence and Pattern of Learning Disability in India: Systematic Review and Meta-Analysis. *Indian Journal of Psychiatric Nursing*, 19(2).
- Kay, K., & Shipman, C. (2014). *The Confidence Code*. HarperCollins.
- Kearney, C.A. (2001). *School Refusal Behavior in Youth: A Functional Approach to Assessment and Treatment*. American Psychological Association.
- Kearney, C.A., & Albano, A.M. (2004). The Functional Profiles of School Refusal Behavior. *Behavior Modification*, 28(1).
- Kessler, R.C. et al. (2003). The Epidemiology of Major Depressive Disorder. *JAMA*, 289(23).
- Klingberg, T. et al. (2005). Computerized Training of Working Memory in Children with ADHD. *Journal of the American Academy of Child & Adolescent Psychiatry*, 44(2).
- Kovacs, M. (1985). The Children's Depression Inventory (CDI). *Psychopharmacology Bulletin*, 21(4).
- Lally, P., van Jaarsveld, C.H.M., Potts, H.W.W., & Wardle, J. (2010). How Are Habits Formed: Modelling Habit Formation in the Real World. *European Journal of Social Psychology*, 40(6).
- Laney, M.O. (2002). *The Introvert Advantage: How to Thrive in an Extrovert World*. Workman Publishing.
- Leonard, L.B. (1998). *Children with Specific Language Impairment*. MIT Press.
- Levelt, W.J.M. (1989). *Speaking: From Intention to Articulation*. MIT Press.
- Linehan, M.M. (1993). *Cognitive-Behavioral Treatment of Borderline Personality Disorder*. Guilford Press.
- Logan, G.D. (1994). On the Ability to Inhibit Thought and Action. *Inhibitory Processes in Attention, Memory, and Language*.
- McHolm, A.E., Cunningham, C.E., & Vanier, M.K. (2005). *Helping Your Child with Selective Mutism*. New Harbinger.
- Ministry of Education, Government of India (2020). *National Education Policy 2020*.
- Ministry of Education, Government of India (2021). *NIPUN Bharat: National Initiative for Proficiency in Reading with Understanding and Numeracy*.
- Miller, L.J. et al. (2007). Concept Evolution in Sensory Integration. *American Journal of Occupational Therapy*, 61(2).
- Neff, K.D. (2011). *Self-Compassion: The Proven Power of Being Kind to Yourself*. William Morrow.
- Nickels, L. (2002). Therapy for Naming Disorders: Revisiting, Revising, and Reviewing. *Aphasiology*, 16(10-11).
- O'Nions, E., Christie, P., Gould, J., Viding, E., & Happe, F. (2014). Development of the 'Extreme Demand Avoidance Questionnaire' (EDA-Q). *Journal of Child Psychology and Psychiatry*, 55(4).
- Pamungkas et al. (2025). Recent Advances in Artificial Intelligence for Dyslexia Detection: A Systematic Review. *International Journal of Robotics and Control Systems*.
- PMC (2024). Bridging the Gap in Adult Dyslexia Research: Assessing the Efficacy of a Linguistic Intervention on Literacy Skills in Adults Aged 16-30.
- Polatajko, H.J., & Cantin, N. (2006). Developmental Coordination Disorder (Dyspraxia): An Overview of the State of the Art. *Seminars in Pediatric Neurology*, 12(4).
- Przybylski, A.K., Murayama, K., DeHaan, C.R., & Gladwell, V. (2013). Motivational, Emotional, and Behavioral Correlates of Fear of Missing Out. *Computers in Human Behavior*, 29(4).

- Raatikainen, P., Hautala, J., Loberg, O., Karkkainen, T., Leppanen, P., & Nieminen, P. (2021). Detection of Developmental Dyslexia with Machine Learning Using Eye Movement Data. *Array*, 12.
- Rakos, R.F. (1991). *Assertive Behavior: Theory, Research, and Training*. Routledge.
- Rangasrinivasan, S. et al. (2025). AI-Enhanced Child Handwriting Analysis: A Framework for the Early Screening of Dyslexia and Dysgraphia. *SN Computer Science*, 6(5).
- Rapee, R.M., & Heimberg, R.G. (1997). A Cognitive-Behavioral Model of Anxiety in Social Phobia. *Behaviour Research and Therapy*, 35(8).
- Reis, S.M., & McCoach, D.B. (2000). The Underachievement of Gifted Students. *Gifted Child Quarterly*, 44(3).
- Sakulku, J. (2011). The Impostor Phenomenon. *The Journal of Behavioral Science*, 6(1).
- Sedmidubsky, J., Kuzik, O., & Zezula, P. (2024). ETDD70: Eye-Tracking Dataset for Classification of Dyslexia Using AI-Based Methods. *Lecture Notes in Computer Science*.
- Shanahan, M.A., Pennington, B.F., Yerys, B.E., et al. (2006). Processing Speed Deficits in Attention Deficit/Hyperactivity Disorder and Reading Disability. *Journal of Abnormal Child Psychology*, 34(5).
- Snowling, M.J., & Hulme, C. (2011). Evidence-Based Interventions for Reading and Language Difficulties. *Journal of Research in Reading*, 34(1).
- St. Louis, K.O., Myers, F.L., Bakker, K., & Raphael, L.J. (2007). Understanding and Treating Cluttering. In *Stuttering and Related Disorders of Fluency*.
- Tangney, J.P., Wagner, P., Fletcher, C., & Gramzow, R. (1992). Shamed into Anger? *Journal of Personality and Social Psychology*, 62(4).
- Thomas, R., Sanders, S., Doust, J., Beller, E., & Glasziou, P. (2015). Prevalence of ADHD: Systematic Review and Meta-Analysis. *Pediatrics*, 135(4).
- Tomblin, J.B., Records, N.L., Buckwalter, P., Zhang, X., Smith, E., & O'Brien, M. (1997). Prevalence of Specific Language Impairment in Kindergarten Children. *Journal of Speech, Language, and Hearing Research*, 40(6).
- Tylka, T.L. (2011). Positive Psychology Perspectives on Body Image. *Body Image: A Handbook of Science, Practice, and Prevention* (2nd ed.). Guilford Press.
- Vajs, I., Savic, A.M., Milanovic, S., Tasic, J., & Ilic, I. (2023). Accessible Dyslexia Detection with Real-Time Reading Feedback. *Brain Sciences*, 13(3).
- van der Kolk, B. (2014). *The Body Keeps the Score*. Viking.
- Vellutino, F.R., Fletcher, J.M., Snowling, M.J., & Scanlon, D.M. (2004). Specific Reading Disability (Dyslexia): What Have We Learned in the Past Four Decades? *Journal of Child Psychology and Psychiatry*, 45(1).
- Vogel, E.A., Rose, J.P., Roberts, L.R., & Eckles, K. (2014). Social Comparison, Social Media, and Self-Evaluation. *Psychology of Popular Media Culture*, 3(4).
- Webb, J.T., Amend, E.R., Webb, N.E., Goerss, J., Beljan, P., & Olenchak, F.R. (2005). *Misdiagnosis and Dual Diagnoses of Gifted Children and Adults*. Great Potential Press.
- WHO / UNICEF (2023). *Global Report on Children with Developmental Disabilities*.
- Williams, K.D. (2007). Ostracism: The Kiss of Social Death. *Social and Personality Psychology Compass*, 1(1).
- Willcutt, E.G., Doyle, A.E., Nigg, J.T., Faraone, S.V., & Pennington, B.F. (2005). Validity of the Executive Function Theory of ADHD. *Biological Psychiatry*, 57(11).
- Wilson, A.J., Revkin, S.K., Cohen, D., Cohen, L., & Dehaene, S. (2006). An Open Trial Assessment of 'The Number Race'. *Behavioral and Brain Functions*, 2(1).
- Young, J.E., Klosko, J.S., & Weishaar, M.E. (2003). *Schema Therapy: A Practitioner's Guide*. Guilford Press.

- Zaibi, T., & Bezine, H. (2024). Early Detection of Learning Disabilities through Handwriting Analysis and Machine Learning. *Procedia Computer Science*, 246, 3702-3712.
- Zeidner, M. (1998). *Test Anxiety: The State of the Art*. Plenum Press.
- Zimbardo, P.G. (1977). *Shyness: What It Is, What to Do About It*. Addison-Wesley.
- Zoogman, S., Goldberg, S.B., Hoyt, W.T., & Miller, L. (2015). Mindfulness Interventions with Youth: A Meta-Analysis. *Mindfulness*, 6(2).

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